

UFA, Russian Federation

WMO: 287220

Lat: 54.7109N

Lon: 55.8139E

Elev: 104

StdP: 100.08

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.9	-27.8	-34.0	0.2	-30.8	-30.6	0.2	-27.6	11.9	-5.8	10.6	-4.8	1.5	230	0.603

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.2	31.8	20.5	29.9	20.0	28.0	19.0	21.9	28.8	21.0	27.8	20.0	26.3	3.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.5	14.4	25.2	18.6	13.6	24.3	17.8	12.9	23.3	64.6	28.9	61.1	27.6	57.8	26.3	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.0	8.6	7.5	-35.1	34.3	3.3	1.8	-37.6	35.6	-39.5	36.7	-41.4	37.7	-43.8	39.0		
(5)			-35.3	24.1	3.3	1.9	-37.6	25.5	-39.6	26.6	-41.4	27.7	-43.8	29.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.0	-13.1	-11.9	-4.7	5.5	13.4	17.7	19.7	17.9	11.9	4.8	-3.5	-10.6		
	DBStd	13.06	7.68	7.22	5.38	5.45	4.80	4.20	3.46	4.06	4.27	4.55	6.29	7.77		
	HDD10.0	3207	717	614	456	156	22	2	0	1	27	169	405	638		
	HDD18.3	5401	975	847	714	385	166	62	27	59	196	418	655	897		
	CDD10.0	1021	0	0	0	21	126	233	300	246	86	9	0	0		
	CDD18.3	172	0	0	0	1	11	42	68	46	4	0	0	0		
	CDH23.3	2058	0	0	0	12	210	508	755	511	62	0	0	0		
	CDH26.7	681	0	0	0	1	51	170	256	189	13	0	0	0		
(14) Wind	WSAvg	3.1	3.3	3.3	3.6	3.4	3.2	2.8	2.5	2.6	2.8	3.3	3.4	3.4		
(15) Precipitation	PrecAvg	578	44	38	37	31	51	62	49	55	47	59	52	53		
	PrecMax	766	91	79	75	79	107	126	154	106	123	114	124	117		
	PrecMin	399	7	7	2	1	4	8	10	7	6	10	3	6		
	PrecStd	100	22	18	19	20	29	33	29	25	28	30	29	26		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.8	2.7	9.2	24.9	30.1	33.0	33.8	34.0	28.7	20.2	10.0	2.4		
		MCWB	1.0	1.1	4.9	14.0	17.7	20.4	21.2	20.5	16.9	11.9	8.4	1.1		
	2%	DB	0.3	1.1	5.5	21.3	28.1	30.9	31.2	31.2	24.9	17.1	7.1	1.4		
		MCWB	-0.3	0.2	2.5	12.4	16.8	20.2	21.1	20.2	15.9	10.6	5.9	0.7		
	5%	DB	-0.9	-0.1	3.6	18.3	25.9	28.7	29.6	28.8	22.0	14.0	5.2	0.8		
		MCWB	-1.6	-0.9	1.4	10.7	15.8	19.4	20.6	19.7	14.9	9.5	4.0	0.2		
	10%	DB	-2.2	-1.9	2.1	15.0	23.0	26.2	27.8	26.0	19.6	11.5	3.5	-0.7		
		MCWB	-2.9	-2.7	0.6	9.1	14.4	18.4	19.8	18.6	13.8	8.5	2.5	-1.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.0	1.3	5.5	14.9	19.5	22.6	23.4	22.7	18.9	13.5	8.7	1.5		
		MCDB	1.4	2.0	8.6	23.7	27.7	29.9	30.4	30.0	25.7	17.9	9.8	1.8		
	2%	WB	-0.2	0.3	2.8	12.9	17.8	21.2	22.3	21.2	17.0	11.6	6.0	0.8		
		MCDB	0.2	0.9	4.8	20.0	25.8	28.2	29.1	28.6	22.7	15.6	6.7	1.2		
	5%	WB	-1.6	-1.0	1.7	11.3	16.5	20.1	21.2	20.2	15.6	10.0	4.1	0.0		
		MCDB	-1.0	-0.2	3.0	17.5	23.8	26.7	27.7	27.0	20.5	12.9	5.0	0.4		
	10%	WB	-3.1	-2.7	0.8	9.6	15.2	19.0	20.3	19.1	14.4	8.7	2.5	-1.3		
		MCDB	-2.4	-1.9	1.9	14.9	21.9	25.2	26.4	25.2	18.5	11.3	3.3	-0.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.3	10.0	10.1	11.1	12.9	12.2	12.2	11.9	10.8	7.7	6.1	7.4		
		MCDBR	6.4	7.4	9.8	16.1	17.3	15.9	15.6	16.1	15.3	13.3	6.1	5.4		
		MCWBR	6.2	6.6	7.1	9.0	8.0	7.2	7.0	7.3	7.8	8.0	5.3	5.2		
	5% WB	MCDBR	6.3	6.6	7.9	15.4	15.2	14.1	14.0	14.5	13.3	11.4	5.6	5.6		
		MCWBR	6.3	6.0	6.0	8.9	7.9	7.0	6.8	7.1	7.4	7.5	5.1	5.5		
(40) Clear-Sky Solar Irradiance	taub		0.303	0.327	0.380	0.422	0.408	0.401	0.418	0.439	0.390	0.362	0.327	0.305		
	taud		2.024	2.002	1.927	2.057	2.245	2.322	2.279	2.196	2.297	2.263	2.159	1.996		
	Ebn at Noon		586	717	762	793	836	846	822	773	765	688	565	480		
	Edh at Noon		72	105	143	145	127	119	122	126	100	82	63	60		
(44) All-Sky Solar Radiation	RadAvg		0.86	1.88	3.32	4.38	5.34	5.82	5.75	4.53	2.89	1.33	0.61	0.52		
	RadStd		0.10	0.18	0.30	0.49	0.50	0.64	0.55	0.46	0.31	0.30	0.14	0.12		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.51	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page